

Yifan Wu

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EDUCATIONAL BACKGROUND

University of Pennsylvania	Philadelphia, USA
School of Engineering and Applied Science	Sept.2025-present
Master of Electrical and Computer Engineering	
ShanghaiTech University	Shanghai, China
School of Information Science and Technology	Sept.2020-Jun.2024
Bachelor of Engineering in Electronic and Information Engineering	

RESEARCH INTERESTS

Tactile Perception and Real-Time Estimation for Contact-Rich Manipulation: fabricating visuotactile sensor hardware, reconstructing contact geometry from it, and building uncertainty-aware estimators and controllers that let a robot decide when partial, noisy contact information is enough to act. Interested in extending this to mobile manipulation and imitation learning in unstructured, contact-heavy environments such as construction and field robotics, where perception is unreliable and contact is unavoidable.

PUBLICATIONS

- [1] Yifan Wu, Yuzhuo Chen, Zhengying Zhu, Xuhao Qin, and Chenxi Xiao, “HumanFT: A Human-like Fingertip Multimodal Visuo-Tactile Sensor”, Published on *ICRA 2025*. [\[Link\]](#)
- [2] Yining Wang, Fengxiang Tang, Boya Liu, Yifan Wu, Ruohan Zhang, Hao Ren, “A bioinspired microbial taste chip with artificial intelligence-enabled high selectivity and ultra-short response time”, Published on *Biosensors and Bioelectronics*. [\[Link\]](#)
- [3] Yining Wang, Fengxiang Tang, Ruohan Zhang, Yifan Wu, and Hao Ren, “A Wireless Mems Microbial Fuel Cell Biosensor with Record Ultra-Short Response time for Copper Ion in Water Monitoring”, Published on *IEEE MEMS 2024*. [\[Link\]](#)
- [4] Peng, Chuhan; Zhou, Mingde; Wu, Yifan; Fu, Minfan; Wang, Haoyu, “Adaptive Virtual Bus Strategy for Electrolytic Capacitorless DAB Microinverters”, Published on *IEEE Transactions on Industrial Electronics*. [\[Link\]](#)

RESEARCH EXPERIENCE

GRASP Laboratory, University of Pennsylvania	Oct.2025–Present
Supervised by Prof. Nadia Figueroa Graduate Research Assistant	
Project Title: AMMH-Grasp: Tactile Active Exploration and Grasping of Unknown Objects	May.2026–Present
- Built a bimanual simulation pipeline where one arm reconstructs an unknown object’s surface purely through touch while the other plans and executes a grasp from the evolving model; - Designed an uncertainty-aware stopping criterion and grasp-candidate selection strategy, showing reliable grasps from partial models without full exploration.	
Project Title: Real-Time Roughness Sensing for Robotic Polishing	Oct.2025–Present
- Built a multi-modal sensor rig and labeled dataset spanning a wide range of abrasive grit levels for closed-loop robotic polishing; - Developed a real-time roughness estimation pipeline and identified where offline benchmarks failed to predict real-world, real-time performance.	
ShanghaiTech Robot Interaction and Manipulation Lab	Sept.2023-Aug.2024
Supervised by Prof. Chenxi Xiao Undergraduate Research Assistant	
Project Title: HumanFT: A Human-like Fingertip Multimodal Visuo-Tactile Sensor	[Link]
Fabricated a compact (12 × 20 × 35 mm) multimodal fingertip sensor: custom PDMS elastomer, an integrated PCB sensing three-axis contact force and vibration, and a photometric-stereo + MLP pipeline for 3D contact reconstruction.	
Power Electronics and Renewable Energies Laboratory, ShanghaiTech	Sept.2024 – Oct.2025
Supervised by Prof. Haoyu Wang Graduate Research Assistant	
Project Title: Adaptive Virtual Bus Strategy for Electrolytic Capacitorless DAB Microinverters	[Link]
Designed and validated a 200 W quasi-single-stage DAB microinverter prototype eliminating electrolytic capacitors via active power decoupling, reaching 92.67% peak measured efficiency.	
Shanghai Engineering Research Center of Energy Efficient and Custom AI IC	Feb.2023-Aug.2023
Supervised by Prof. Hao Ren Undergraduate Research Assistant	
Project Title: Real-time Wireless Microfluidic Microbial Fuel Cell Biosensor	[Link 1] [Link 2]
Designed a wireless microfluidic biosensor for copper-ion detection with a validated kinetic model, improving response and recovery time by 3.75× over prior designs.	

INTERNSHIP

Baidu China	Jan.2024-Feb.2024
Technical Intern Service Experience Product R&D Team – large-scale multimodal LLM testing.	

PROFESSIONAL SKILLS

Software: C/C++, Python (NumPy, SciPy), MATLAB/Simulink, MuJoCo, ROS, PyTorch, SolidWorks, Altium

Hardware: PDMS elastomer casting, photometric stereo, PCB design, STM32CubeIDE/HAL, ESP-IDF, I2C/SPI/UART